



JOHNS HOPKINS

WHITING SCHOOL
of ENGINEERING

MECHANICAL ENGINEERING COURSE PLANNING MAPS

Note: This document is intended to serve as a scheduling aid for Mechanical Engineering students. The schedule information provided is based on historical patterns and expected scheduling. Course offerings and modalities are subject to change due to instructor availability, student demand, and program priority. Students should check SIS for official course schedule information and the Johns Hopkins University Academic Catalog for official degree requirements.

FOCUS AREAS

Click to navigate to your focus area.

ADVANCED MANUFACTURING

HYPERSONIC TECHNOLOGIES

AEROSPACE ENGINEERING

MECHANICS OF MATERIALS AND
STRUCTURES

BIOMECHANICS

OCEAN ENGINEERING

FLUID AND THERMAL MECHANICS

ROBOTICS, DYNAMICS, AND
CONTROLS

ADVANCED MANUFACTURING

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.628
Computer-Integrated Design &
Manufacturing

Every Term

535.642
Control Systems for Mechanical
Engineering Applications

Fall

535.659
Manufacturing System Analysis

Fall

535.660
Precision Mechanical Design

Fall/Spring

535.673
Mechanized Assembly: Hardware
and Algorithms

Summer

535.674
Smart Manufacturing and
Automation

Spring

535.721
Advanced Composite Materials &
Manufacturing Processes

Spring/Summer

Group 2 Courses

535.737
Multiscale Modeling & Simulation
of Mechanical Systems

Summer

535.684
Modern Polymeric Materials

Fall/Spring

535.720
Mechanics of Composite
Materials and Structures

Fall

535.627
Computer-Aided Design

Every Term

535.744
AI for Mechanical Materials
Design

Spring

Computational

Materials

535.618
Fabricatology: Advanced
Materials Processing

Summer/Fall

535.638
Mechanical Packaging for
Electronics Systems

Summer/Fall

535.733
Micromechanics of
Heterogeneous Materials &
Composites

Fall

535.672
Advanced Manufacturing
Systems

Summer

515.658
Design for Additive
Manufacturing

Fall

515.622
Micro and Nano Structured
Materials & Devices

TBD

Applications

AEROSPACE ENGINEERING [AERODYNAMICS FOCUS AREA]

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.608
Hypersonic Technologies and
Systems

Fall

535.620
Fluid Dynamics I

Summer/Fall

535.626
Mechanics of Flight

Fall/Spring

535.670
Advanced Aerodynamics

Fall

Group 2 Courses

535.608
Hypersonic Technologies and
Systems

Fall

535.623
Intermediate Vibrations

Fall/Spring

535.625
Turbulence

Fall

535.626
Mechanics of Flight

Fall/Spring

535.639
Aerospace Materials

Spring

535.652
Thermal Systems Design and
Analysis

Fall

535.670
Advanced Aerodynamics

Fall

535.735
Computational Fluid Dynamics

Fall

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles

Fall

535.761
Hypersonic Aerothermodynamics

Spring

535.763
Aerospace Propulsion

Spring

AEROSPACE ENGINEERING [DYNAMICS & CONTROL FOCUS AREA]

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.608
Hypersonic Technologies and
Systems

Fall

535.626
Mechanics of Flight

Fall/Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles

Fall

535.762
Guidance, Navigation and
Controls for Hypersonic Vehicles

Fall/Spring

Group 2 Courses

535.608
Hypersonic Technologies and
Systems

Fall

535.612
Intermediate Dynamics

Fall

535.623
Intermediate Vibrations

Fall/Spring

535.626
Mechanics of Flight

Fall/Spring

535.642
Control Systems for Mechanical
Engineering Applications

Fall

535.670
Advanced Aerodynamics

Fall

535.724
Dynamics of Robots and
Spacecraft

Spring

535.741
Optimal Control and
Reinforcement Learning

Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles

Fall

535.762
Guidance, Navigation and
Controls for Hypersonic Vehicles

Fall/Spring

535.763
Aerospace Propulsion

Spring

525.609
Continuous Control Systems

Fall/Spring

525.645
Modern Navigation Systems

Fall/Spring

525.661
UAV Systems and Controls

Fall/Spring

525.777
Control System Design Methods

Fall

AEROSPACE ENGINEERING [STRUCTURES & MATERIALS FOCUS AREA]

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.606
Advanced Strength of Materials
Spring

535.607
Mechanics of Solids & Structures:
Theory and Applications I
Fall

535.632
Applied Finite Elements
Every Term

Group 2 Courses

535.606
Advanced Strength of Materials
Spring

535.607
Mechanics of Solids & Structures:
Theory and Applications I
Fall

535.608
Hypersonic Technologies and
Systems
Fall

535.612
Intermediate Dynamics
Fall

535.623
Intermediate Vibrations
Fall/Spring

535.627
Computer-Aided Design
Every Term

535.628
Computer-Integrated Design &
Manufacturing
Every Term

535.632
Applied Finite Elements
Every Term

535.639
Aerospace Materials
Spring

535.643
Plasticity
Spring

535.652
Thermal Systems Design and
Analysis
Fall

535.660
Precision Mechanical Design
Fall/Spring

535.684
Modern Polymeric Materials
Fall/Spring

535.706
Mechanics of Solids & Structures:
Theory and Applications II
Spring

535.720
Mechanics of Composite
Materials and Structures
Fall

535.721
Advanced Composite Materials &
Manufacturing Processes
Spring/Summer

535.731
Engineering Materials: Properties
and Selections
Fall/Spring

535.732
Fatigue and Fracture of Materials
Spring

515.658
Design for Additive
Manufacturing
Fall

515.661
Introduction to Polymer Science
Fall

675.702
Spacecraft Materials
Every Term

675.722
Space Mechanical Design and
Analysis
In Development

AEROSPACE ENGINEERING [VEHICLE DESIGN FOCUS AREA]

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.608
Hypersonic Technologies and
Systems

Fall

535.626
Mechanics of Flight

Fall/Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles

Fall

Group 2 Courses

535.606
Advanced Strength of Materials

Spring

535.607
Mechanics of Solids & Structures:
Theory and Applications I

Fall

535.608
Hypersonic Technologies and
Systems

Fall

535.626
Mechanics of Flight

Fall/Spring

535.627
Computer-Aided Design

Every Term

535.628
Computer-Integrated Design &
Manufacturing

Every Term

535.632
Applied Finite Elements

Every Term

535.652
Thermal Systems Design and
Analysis

Fall

535.660
Precision Mechanical Design

Fall/Spring

535.670
Advanced Aerodynamics

Fall

535.706
Mechanics of Solids & Structures:
Theory and Applications II

Spring

535.720
Mechanics of Composite
Materials and Structures

Fall

535.721
Advanced Composite Materials &
Manufacturing Processes

Spring/Summer

535.724
Dynamics of Robots and
Spacecrafts

Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles

Fall

535.761
Hypersonic Aerothermodynamics

Spring

535.763
Aerospace Propulsion

Spring

675.722
Space Mechanical Design and
Analysis

In Development

BIOMECHANICS

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.661
Biofluid Mechanics

Fall

535.663
Biosolid Mechanics

Spring/Summer

585.631
Introduction to Biomechanics

Spring

Group 2 Courses

535.664
Fundamental Principles for Bio-
microfluidic Systems

Spring

535.667
Biomechanics of Human
Movement

Summer/Fall

535.693
Fabrication of Biomaterials,
Engineering Tissues, and Food

Fall

585.612
Biomechanical Engineering in
Sports

Fall

585.729
Cell and Tissue Engineering

Fall

585.742
Regenerative Tissue Engineering

Spring

585.743
Modeling Approaches to Cell
and Tissue Engineering

Spring

585.747
Advances in Cardiovascular
Medicine

Fall

585.762
Computational Biomechanics
and Physiological Modeling

Fall

Human Applications

535.691
Haptic Interface Design

Fall

525.786
Human Robotics Interaction

Spring

Human Robots

585.720
Orthopedic Biomechanics

Fall/Spring

535.750
Biomechanics of the cell: From
nano- and micro-mechanics to
cell organization & function

Fall

585.601
Physiology for Applied
Biomedical Engineering I

Fall/Spring

585.708
Biomaterials

Summer

585.709
Biomechanics of Cells and Stem
Cells

Fall

585.710
Biochemical Sensors

Fall/Spring

585.718
Biological Solid & Fluid
Mechanics

Summer

Other Applications

585.726
Biomimetics in Biomedical
Engineering

Summer

FLUID AND THERMAL MECHANICS

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.620
Fluid Dynamics I

Summer/Fall

535.633
Intermediate Heat Transfer

Spring

Group 2 Courses

535.625
Turbulence

Fall

535.626
Mechanics of Flight

Fall/Spring

535.614
Fundamentals of Acoustics

Spring

535.771
Naval Architecture Design

Spring

535.670
Advanced Aerodynamics

Fall

535.761
Hypersonic Aerothermodynamics

Spring

535.773
Acoustical Oceanography

Fall

615.761
Intro to Oceanography

Fall

535.763
Aerospace Propulsion

Spring

Aerospace
Applications

Ocean
Applications

535.661
Biofluid Mechanics

Fall

535.664
Fundamental Principles for Bio-
microfluidic Systems

Spring

Bio
Applications

535.634
Applied Heat Transfer

Summer

535.652
Thermal Systems Design and
Analysis

Fall

535.662
Energy and Environment

Spring

535.735
Computational Fluid Dynamics

Fall

535.737
Multiscale Modeling and
Simulation of Mechanical
Systems

Summer

Other
Applications

HYPERSONIC TECHNOLOGIES

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.608
Hypersonic Technologies and
Systems

Fall

535.734
Ultra-high Temperature Materials

Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles

Fall

535.761
Hypersonic Aerothermodynamics

Spring

Group 2 Courses

535.626
Mechanics of Flight

Fall/Spring

535.633
Intermediate Heat Transfer

Spring

535.634
Applied Heat Transfer

Summer

535.670
Advanced Aerodynamics

Fall

535.735
Computational Fluid Dynamics

Fall

Fluids/Thermal

535.671
Aerospace Materials, Structures
and Design

Fall/Spring

535.721
Advanced Composite Materials &
Manufacturing Processes

Spring

Solids and Structures

535.741
Optimal Control and
Reinforcement Learning

Spring

535.762
Guidance, Navigation and
Controls for Hypersonic Vehicles

Fall/Spring

Robotics

535.627
Computer-Aided Design

Every Term

Other
Applications

535.763
Aerospace Propulsion

Spring

Propulsion

MECHANICS OF MATERIALS AND STRUCTURES

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.632
Applied Finite Elements

Every Term

CAD/FEA

535.731
Engineering Materials: Properties
and Selections

Fall/Spring

Materials Science

535.623
Intermediate Vibrations

Fall/Spring

Dynamics/Controls

535.606
Advanced Strength of Materials

Spring

535.607
Mechanics of Solids & Structures:
Theory and Applications I

Fall

Mechanics

MECHANICS OF MATERIALS AND STRUCTURES

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 2 Courses

535.627 Computer-Aided Design Every Term CAD	535.706 Mechanics of Solids & Structures: Theory and Applications II Spring	535.660 Precision Mechanical Design Fall/Spring	535.720 Mechanics of Composite Materials and Structures Fall	535.618 Fabricatology: Advanced Materials Processing Summer/Fall	515.601 Structure and Properties of Materials Fall
535.663 Biosolid Mechanics Spring/Summer	535.732 Fatigue and Fracture of Materials Spring	565.682 Design of Ocean Structures Summer (Odd years)	535.721 Advanced Composite Materials & Manufacturing Processes Spring/Summer	535.674 Smart Manufacturing and Automation Spring	515.617 Nanomaterials Spring/Summer
585.631 Introduction to Biomechanics Spring Biomechanics	535.748 Stress Waves, Impacts and Shockwaves Spring	675.722 Space Mechanical Systems Design and Analysis In Development	535.733 Micromechanics of Heterogeneous Materials & Composites Fall	515.658 Design for Additive Manufacturing Fall	515.622 Micro and Nano Structured Materials & Devices TBD
535.612 Intermediate Dynamics Fall	535.643 Plasticity Spring	Machine & Structure Design	Composites	Additive	525.606 Electronic Materials In Development
585.631 Introduction to Biomechanics Spring Dynamics	565.604 Structural Mechanics Fall	515.602 Thermodynamics and Kinetics of Materials Spring	535.684 Modern Polymeric Materials Summer/Fall	535.742 Applied Machine Learning for Mechanical Engineers Spring/Summer	675.702 Materials for Space Systems Every Term
	Mechanics	Thermodynamic	515.661 Introduction to Polymer Science Fall	535.743 Intermediate Applied AI in Mechanical Engineering Fall	Materials
		Electronic Materials	545.660 Polymer Physics In Development	535.744 AI for Mechanical Materials Design Spring	535.737 Multiscale Modeling and Simulation of Mechanical Systems Summer
			Polymer Science	AI/ML	Computation

OCEAN ENGINEERING

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.606
Advanced Strength of Materials

Spring

535.620
Fluid Dynamics I

Summer/Fall

Core ME Classes

535.771
Naval Architecture Design

Summer/Fall

565.682
Design of Ocean Structures

Summer (Odd years)

Marine Vehicles & Naval

535.773
Acoustical Oceanography

Fall

615.761
Intro to Oceanography

Fall

Ocean Sciences

Group 2 Courses

535.607
Mechanics of Solids & Structures:
Theory and Applications I

Fall

535.627
Computer-Aided Design

Every Term

535.632
Applied Finite Elements

Every Term

535.721
Advanced Composite Materials &
Manufacturing Processes

Spring/Summer

535.732
Fatigue and Fracture of Materials

Spring

535.735
Computational Fluid Dynamics

Fall

Core ME Classes

535.638
Mechanical Packaging for
Electronics Systems

Summer/Fall

535.733
Micromechanics of
Heterogeneous Materials &
Composites

Fall

535.614
Fundamentals of Acoustics

Spring

515.658
Design for Additive
Manufacturing

Fall

Marine Vehicles & Naval

615.775
Physics of Climate

Spring

Ocean Sciences

ROBOTICS, DYNAMICS, AND CONTROLS

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.642
Control Systems for Mechanical
Engineering Applications

Fall

Advanced
Control Theory

535.724
Dynamics of Robots and
Spacecraft

Spring

535.630
Kinematics & Dynamics of Robots

Fall/Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles

Fall

Advanced
Control Theory

ROBOTICS, DYNAMICS, AND CONTROLS

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 2 Courses

535.635
Introduction to Mechatronics

Spring

535.673
Mechanized Assembly: Hardware
and Algorithms

Summer

535.691
Haptic Interface Design

Fall

535.782
Haptic Applications

Spring/Summer

525.610
Microprocessors for Robotic
Systems

Spring

525.786
Human Robotics Interaction

Spring

Electronics

535.627
Computer-Aided Design

Every Term

535.612
Intermediate Dynamics

Fall

535.623
Intermediate Vibrations

Fall/Spring

535.630
Kinematics & Dynamics of Robots

Fall/Spring

535.724
Dynamics of Robots and
Spacecraft

Spring

535.628
Computer-Integrated Design &
Manufacturing

Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles

Fall

525.661
UAV Systems and Control

Fall/Spring

605.613
Introduction to Robotics

Fall/Spring

605.716
Modeling and Simulation of
Complex Systems

Spring/Summer

Dynamics

535.638
Mechanical Packaging for
Electronics Systems

Summer/Fall

535.659
Manufacturing Systems Analysis

Fall

535.660
Precision Mechanical Design

Fall/Spring

535.603
Applied Optimal Control

Fall

535.622
Robot Motion Planning

Spring/Summer

535.741
Optimal Control and
Reinforcement Learning

Spring

665.681
Application of Sensing Systems

Every Term

535.642
Control Systems for Mechanical
Engineering Applications

Fall

535.645
Digital Control and Systems
Applications

Summer

525.609
Continuous Control Systems

Fall/Spring

525.626
Feedback Control in Biological
Signaling Pathways

Fall

525.645
Modern Navigation Systems

Fall/Spring

525.777
Control Systems

Fall

Intelligence & Planning

Mechanical Design

Control Theory



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